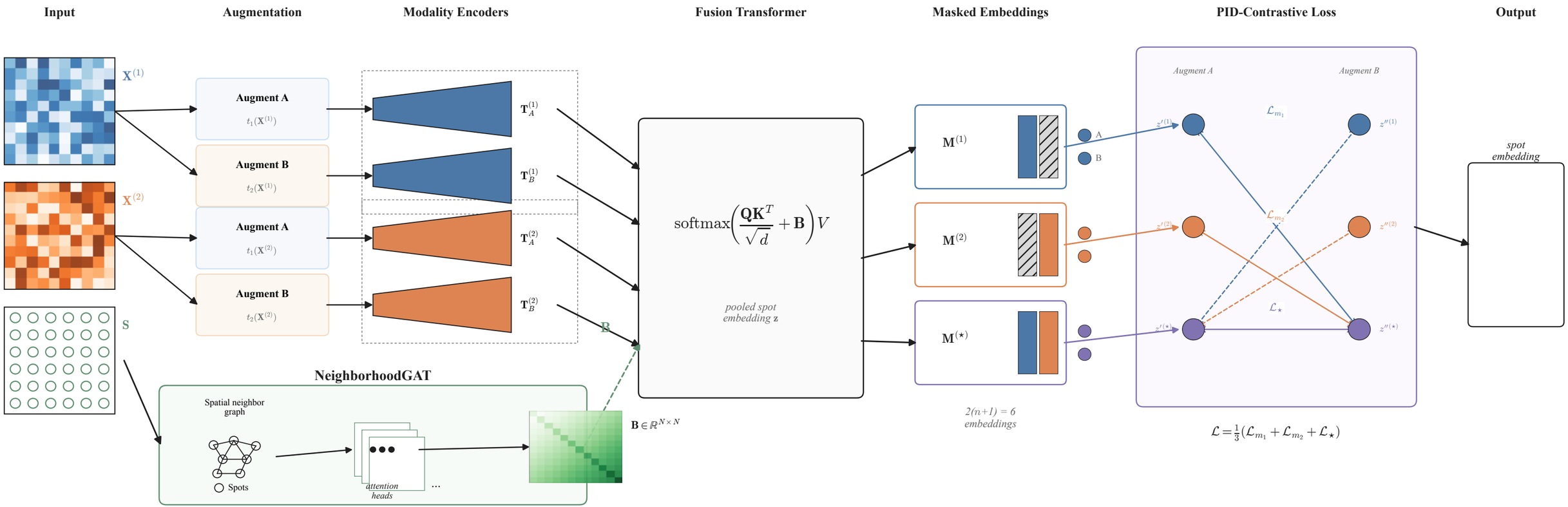

Input
Augmentation
Modality Encoders
Fusion Transformer
Masked Embeddings
PID-Contrastive Loss
Output
$\mathbf{X}^{(1)}$
$\mathbf{X}^{(2)}$
$\mathbf{S}$
Augment A
$t_1(\mathbf{X}^{(1)})$
Augment B
$t_2(\mathbf{X}^{(1)})$
Augment A
$t_1(\mathbf{X}^{(2)})$
Augment B
$t_2(\mathbf{X}^{(2)})$
$\mathbf{T}_A^{(1)}$
$\mathbf{T}_B^{(1)}$
$\mathbf{T}_A^{(2)}$
$\mathbf{T}_B^{(2)}$
$\mathbf{B}$
NeighborhoodGAT
Spatial neighbor graph
Spots
attention heads
...
$\mathbf{B} \in \mathbb{R}^{N \times N}$
$\mathrm{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d}} + \mathbf{B}\right)V$
pooled spot embedding $\mathbf{z}$
$\mathbf{M}^{(1)}$
$\mathbf{M}^{(2)}$
$\mathbf{M}^{(\star)}$
A
B
2(n+1) = 6 embeddings
Augment A
Augment B
$\mathcal{L}_{m_1}$
$\mathcal{L}_{m_2}$
$\mathcal{L}_\star$
$z'^{(1)}$
$z''^{(1)}$
$z'^{(2)}$
$z''^{(2)}$
$z'^{(\star)}$
$z''^{(\star)}$
$\mathcal{L} = \frac{1}{3}(\mathcal{L}_{m_1} + \mathcal{L}_{m_2} + \mathcal{L}_\star)$
spot embedding